

# Fine-Stepping vs EncounterNN Results Report

## *Comparison of EncounterNN Fix 1 against Zone-3-only finer leapfrog stepping*

Prepared from the uploaded evaluation\_encounter\_fine\_stepping.py output files: global, isolated-local, ensemble, and summary verdict reports for  $dt = 0.06, 0.08, \text{ and } 0.10$ .

### 1. Executive Summary

The fine-stepping experiment asks a clean question: at the same global timestep, is it better to spend extra work on EncounterNN or on deterministic smaller leapfrog steps only during the same gated Zone-3 encounter?

The answer is mixed, but not random. EncounterNN clearly learned the local encounter residual: in isolated 0.5-year exit-state tests it gives strong corrections at all three timesteps, especially  $dt=0.08$ . However, the 100-year global rollout is chaotic, so a better local exit state does not always become a better final 100-year trajectory.

At  $dt=0.08$ , EncounterNN is the strongest overall result: it wins the global final-position metric, wins the isolated local exit-state test, and is the ensemble mean winner. Fine stepping at  $k=2$  and  $k=4$  does not reproduce the local benefit and worsens global final position.

At  $dt=0.06$ , EncounterNN wins local exit-state and ensemble mean, but finer  $k4$  has the best single-run global final-position value. This means EncounterNN is locally better but not always globally better in a single chaotic rollout.

At  $dt=0.10$ , finer  $k2$  is best in the single global run, while  $k4$  becomes energy unsafe in the global run. EncounterNN still helps locally, but its global performance is weak at this large timestep. This makes  $dt=0.10$  a caution case rather than a clean EncounterNN success.

Overall conclusion: EncounterNN is not simply beaten by fine stepping. It is a genuine learned local correction, strongest at  $dt=0.08$ , but its long-horizon benefit is inconsistent. Fine stepping is also not a universal replacement, because it can fail or become energy unsafe when the timestep is too large.

### 2. Methodology: What Was Compared

The evaluator compares four SIMON variants against IAS15, which is used as the reference trajectory. noNN is the coarse leapfrog baseline. EncounterNN uses the same coarse leapfrog baseline but, when the pair-1/2 Zone-3 gate fires, it runs the 0.5-year noNN window and applies one learned 18D residual correction at the exit. The Fix 1 rule allows at most one accepted EncounterNN correction per global rollout.

The fine-stepping alternatives do not use a neural network. Instead, when the same pair-1/2 gate fires in Zone 3, the macro-step is split into  $k$  smaller leapfrog substeps.  $k=2$  means the Zone-3 step is halved, and  $k=4$  means it is quartered. The global timestep remains unchanged; only the gated Zone-3 region is refined.

The gate is shared between the neural and fine-stepping methods: pair 1-2 must be in Zone 3 and approaching, with normalized radial velocity below  $-0.40$ . This keeps the comparison focused: NN correction versus extra deterministic stepping in the same physical event region.

**Table 1. Evaluation criteria used in the report.**

| Evaluation level      | What it measures                                                                          | How to read it                                                             |
|-----------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Global 100-year run   | Compares long-horizon accuracy, speedup, and energy drift for one IC1 rollout.            | Useful for real rollout behavior, but sensitive to chaotic branch changes. |
| Isolated local replay | Starts all methods from the same noNN event-entry state and compares against local IAS15. | Best test of whether the method fixes the encounter itself.                |

|                       |                                                                        |                                                                                                          |
|-----------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Ensemble perturbation | Repeats the comparison across 10 tiny initial-condition perturbations. | Tests robustness rather than relying on one chaotic trajectory.                                          |
| Energy safety         | Flags cases with large relative energy drift.                          | Important because a low position error is not reliable if the trajectory becomes numerically unphysical. |

### 3. High-Level Verdict by Timestep

Table 2. Summary verdict across the three timesteps.

| dt   | Global single-run winner          | Isolated exit winner                                 | Ensemble mean winner                            | Interpretation                                                                                           |
|------|-----------------------------------|------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 0.06 | finer-k4 on global final position | EncounterNN                                          | EncounterNN                                     | EncounterNN locally strong; finer stepping slightly better in one global final-position snapshot.        |
| 0.08 | EncounterNN                       | EncounterNN                                          | EncounterNN                                     | Cleanest EncounterNN success case. Fine stepping worsens final position and local exit.                  |
| 0.10 | finer-k2; k4 energy unsafe        | finer-k4 by exit-position, EncounterNN strong second | EncounterNN by mean, but mean is still negative | Large timestep is unstable/mixed. EncounterNN not reliable globally; k4 should be disqualified globally. |

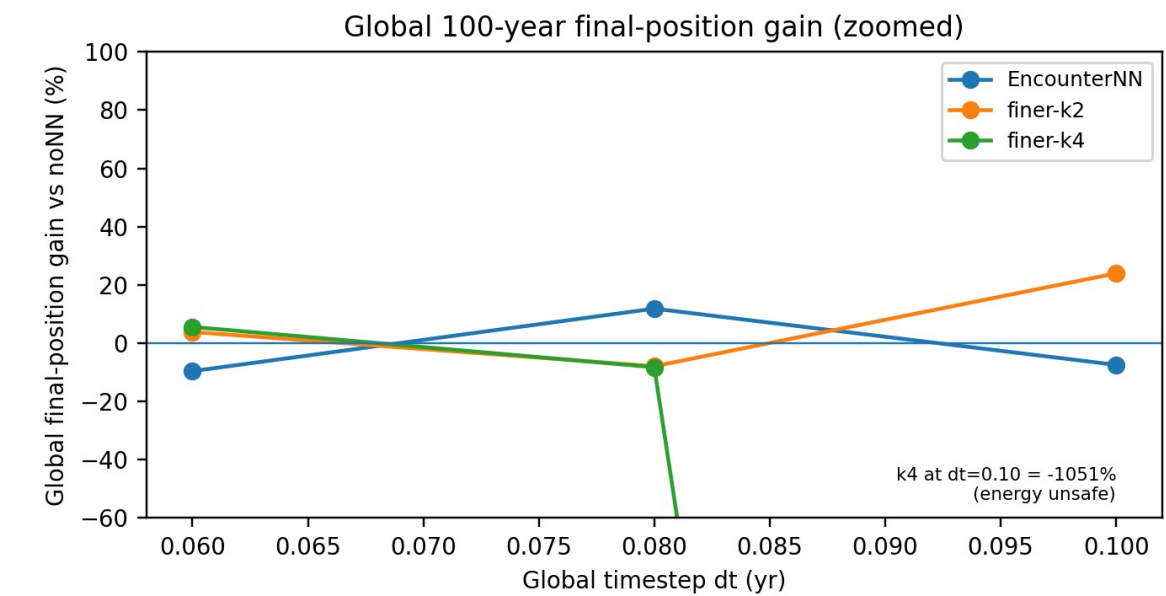


Figure 1. Global final-position gain relative to noNN. The chart is zoomed; the dt=0.10 k4 failure is annotated separately.

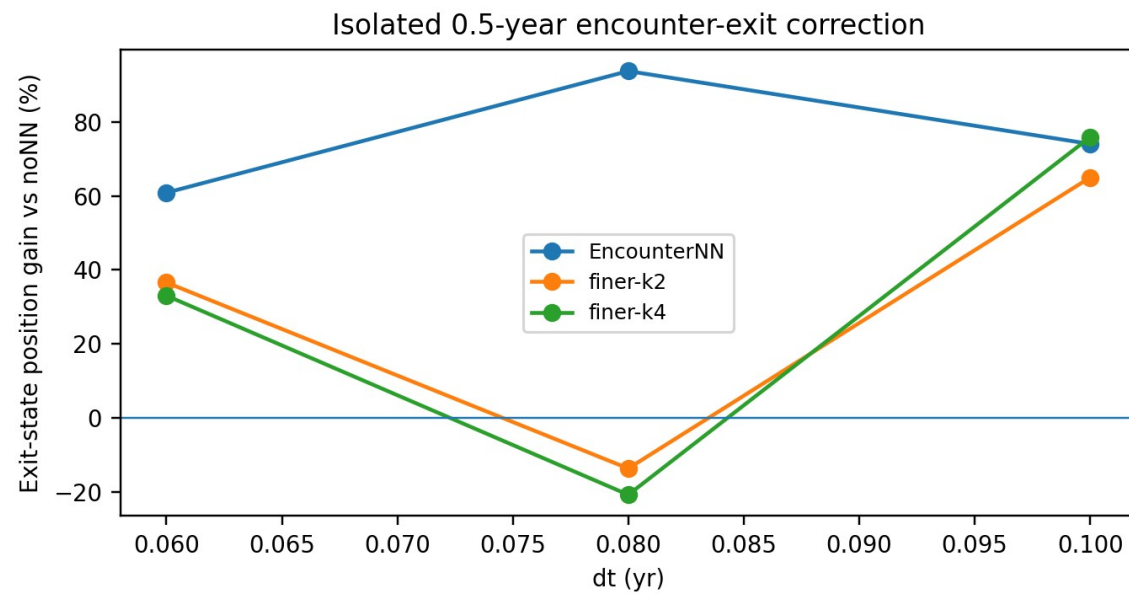


Figure 2. Isolated 0.5-year encounter-exit position gain.

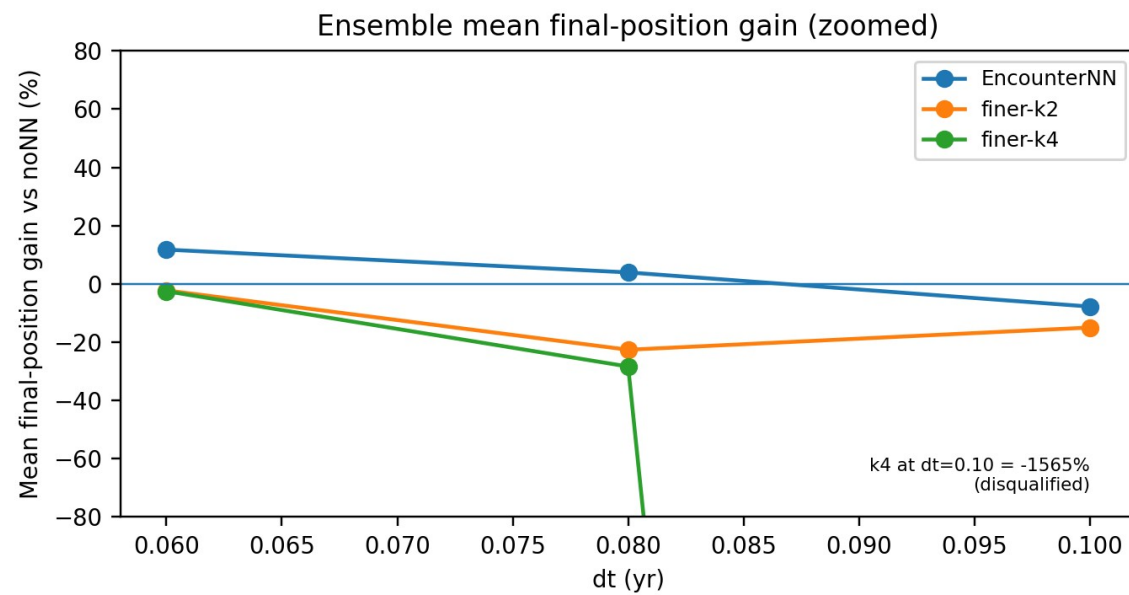


Figure 3. Ensemble mean final-position gain across 10 perturbations. The dt=0.10 k4 failure is annotated separately.

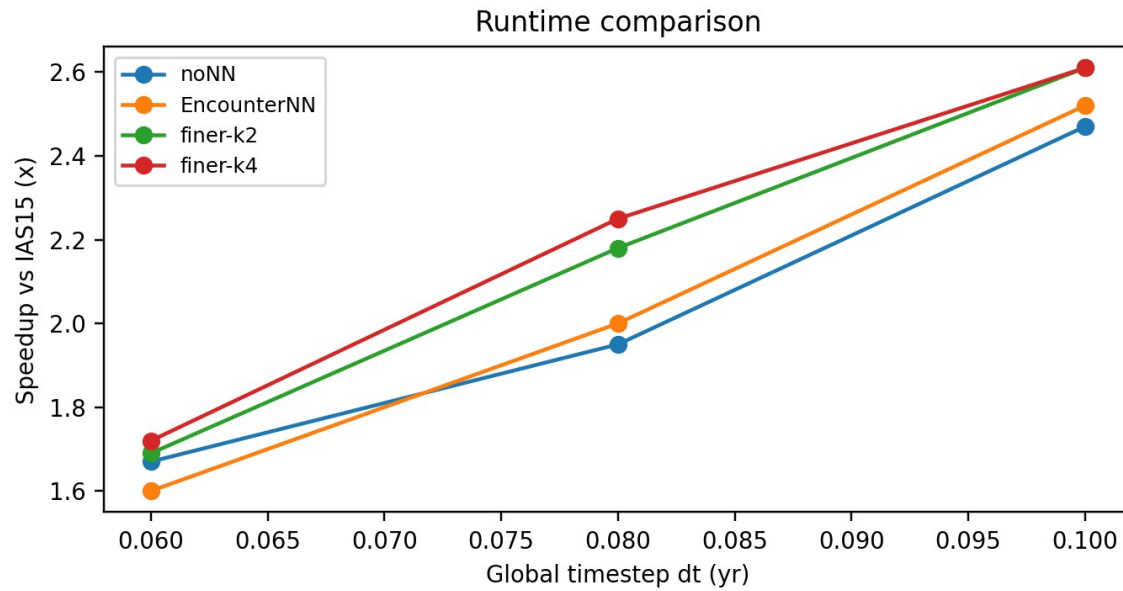


Figure 4. Runtime speedup compared with IAS15.

## 4. Global 100-Year Results

The global run is the most realistic test because it measures the whole 100-year trajectory. However, because this is a chaotic three-body system, a local improvement can still land the trajectory on a different long-term branch. For that reason, global final-position error should be read together with time-average error, energy drift, and the isolated-local test.

Table 3. Global 100-year metrics. Lower error and lower relative energy drift are better; higher speedup is better.

| dt   | method      | pos_final | pos_timeavg | vel_final | vel_timeavg | time_s | speedup | relE     | zone3_fine |
|------|-------------|-----------|-------------|-----------|-------------|--------|---------|----------|------------|
| 0.06 | noNN        | 1.537     | 1.487       | 1.045     | 0.9301      | 0.0890 | 1.67x   | 1.50e-04 | -          |
| 0.06 | EncounterNN | 1.685     | 1.402       | 1.284     | 0.8635      | 0.0927 | 1.60x   | 9.89e-04 | enc=1      |
| 0.06 | finer-k2    | 1.478     | 1.487       | 1.019     | 0.928       | 0.0880 | 1.69x   | 8.03e-04 | 4          |
| 0.06 | finer-k4    | 1.451     | 1.486       | 1.011     | 0.9267      | 0.0865 | 1.72x   | 9.69e-04 | 4          |
| 0.08 | noNN        | 1.624     | 0.9891      | 1.183     | 0.865       | 0.0770 | 1.95x   | 4.49e-04 | -          |
| 0.08 | EncounterNN | 1.431     | 1.008       | 0.6392    | 0.8363      | 0.0751 | 2.00x   | 1.70e-06 | enc=1      |
| 0.08 | finer-k2    | 1.752     | 1.007       | 1.202     | 0.8727      | 0.0689 | 2.18x   | 5.00e-03 | 3          |
| 0.08 | finer-k4    | 1.758     | 1.003       | 1.236     | 0.8762      | 0.0668 | 2.25x   | 6.45e-03 | 3          |
| 0.10 | noNN        | 1.427     | 1.081       | 0.9224    | 0.8951      | 0.0393 | 2.47x   | 5.37e-03 | -          |
| 0.10 | EncounterNN | 1.534     | 1.214       | 1.145     | 0.9825      | 0.0386 | 2.52x   | 2.34e-02 | enc=1      |
| 0.10 | finer-k2    | 1.085     | 1.137       | 0.865     | 0.9362      | 0.0371 | 2.61x   | 5.76e-03 | 4          |
| 0.10 | finer-k4    | 16.43     | 5.232       | 1.036     | 0.8566      | 0.0372 | 2.61x   | 1.62e+00 | 9          |

Table 4. Global percentage gains versus noNN. Positive means the method is closer to IAS15 than noNN.

| dt   | method      | pos_final_gain_pct | pos_timeavg_gain_pct | vel_final_gain_pct | vel_timeavg_gain_pct |
|------|-------------|--------------------|----------------------|--------------------|----------------------|
| 0.06 | EncounterNN | -9.59%             | +5.73%               | -22.84%            | +7.16%               |
| 0.06 | finer-k2    | +3.85%             | -0.01%               | +2.51%             | +0.23%               |

|      |             |           |          |         |        |
|------|-------------|-----------|----------|---------|--------|
| 0.06 | finer-k4    | +5.58%    | +0.05%   | +3.24%  | +0.37% |
| 0.08 | EncounterNN | +11.86%   | -1.91%   | +45.95% | +3.32% |
| 0.08 | finer-k2    | -7.91%    | -1.78%   | -1.65%  | -0.89% |
| 0.08 | finer-k4    | -8.25%    | -1.41%   | -4.51%  | -1.29% |
| 0.10 | EncounterNN | -7.44%    | -12.31%  | -24.11% | -9.77% |
| 0.10 | finer-k2    | +23.97%   | -5.17%   | +6.22%  | -4.59% |
| 0.10 | finer-k4    | -1051.32% | -383.89% | -12.31% | +4.31% |

At dt=0.06, finer-k4 has the best global final-position result: +5.58% versus noNN. EncounterNN worsens final position by -9.59%, but it improves position time-average by +5.73% and velocity time-average by +7.16%. This is a split result: fine stepping wins the final snapshot, while EncounterNN helps the average trajectory and velocity over time.

At dt=0.08, EncounterNN is clearly strongest in the global run. It improves final position by +11.86% and final velocity by +45.95%, with extremely small correction energy error. Both finer-k2 and finer-k4 make the final position worse than noNN.

At dt=0.10, finer-k2 wins the global final-position comparison with +23.97% gain. EncounterNN worsens all four global accuracy metrics. Finer-k4 is not acceptable despite one velocity-timeavg improvement because its relative energy drift is 1.618, which is far beyond the 10% safety threshold.

## 5. Isolated Local Replay Results

The isolated local replay removes pre-encounter drift. All branches start from the same event-entry state, so this is the cleanest test of whether the method actually fixes the close encounter itself.

**Table 5. Isolated local metrics. Positive gain means closer to local IAS15 than noNN.**

| dt   | section     | method      | pos_err   | pos_gain_pct | vel_err   | vel_gain_pct |
|------|-------------|-------------|-----------|--------------|-----------|--------------|
| 0.06 | 0.5yr exit  | noNN        | 0.0008922 | +0.00%       | 0.004263  | +0.00%       |
| 0.06 | 0.5yr exit  | EncounterNN | 0.0003492 | +60.86%      | 0.0007103 | +83.34%      |
| 0.06 | 0.5yr exit  | finer-k2    | 0.0005655 | +36.62%      | 0.004106  | +3.69%       |
| 0.06 | 0.5yr exit  | finer-k4    | 0.0005972 | +33.07%      | 0.004116  | +3.46%       |
| 0.06 | 4yr timeavg | noNN        | 0.0249    | +0.00%       | 0.0291    | -            |
| 0.06 | 4yr timeavg | EncounterNN | 0.0281    | -12.89%      | 0.03239   | -            |
| 0.06 | 4yr timeavg | finer-k2    | 0.02557   | -2.73%       | 0.03123   | -            |
| 0.06 | 4yr timeavg | finer-k4    | 0.02601   | -4.48%       | 0.03188   | -            |
| 0.08 | 0.5yr exit  | noNN        | 0.006271  | +0.00%       | 0.02855   | +0.00%       |
| 0.08 | 0.5yr exit  | EncounterNN | 0.000396  | +93.68%      | 0.001709  | +94.01%      |
| 0.08 | 0.5yr exit  | finer-k2    | 0.007122  | -13.58%      | 0.04048   | -41.77%      |
| 0.08 | 0.5yr exit  | finer-k4    | 0.007567  | -20.67%      | 0.04404   | -54.26%      |
| 0.08 | 4yr timeavg | noNN        | 0.07126   | +0.00%       | 0.06568   | -            |
| 0.08 | 4yr timeavg | EncounterNN | 0.03654   | +48.73%      | 0.039     | -            |
| 0.08 | 4yr timeavg | finer-k2    | 0.09819   | -37.79%      | 0.09851   | -            |
| 0.08 | 4yr timeavg | finer-k4    | 0.1081    | -51.69%      | 0.109     | -            |
| 0.10 | 0.5yr exit  | noNN        | 0.008004  | +0.00%       | 0.1202    | +0.00%       |
| 0.10 | 0.5yr exit  | EncounterNN | 0.002079  | +74.02%      | 0.02712   | +77.43%      |
| 0.10 | 0.5yr exit  | finer-k2    | 0.002805  | +64.96%      | 0.0395    | +67.14%      |
| 0.10 | 0.5yr exit  | finer-k4    | 0.001933  | +75.85%      | 0.03773   | +68.60%      |
| 0.10 | 4yr timeavg | noNN        | 0.2284    | +0.00%       | 0.2399    | -            |
| 0.10 | 4yr timeavg | EncounterNN | 0.1097    | +51.97%      | 0.1688    | -            |
| 0.10 | 4yr timeavg | finer-k2    | 0.09687   | +57.59%      | 0.1716    | -            |
| 0.10 | 4yr timeavg | finer-k4    | 0.1525    | +33.23%      | 0.1885    | -            |

At dt=0.06, EncounterNN is the exit-state winner: position error improves by +60.86% and velocity error by +83.34%. Fine stepping improves position exit error but barely improves velocity exit error. Over the longer 4-year local window, however, all methods are worse than noNN in position time-average, so the local correction does not remain beneficial over that full window.

At  $dt=0.08$ , EncounterNN is the strongest local result by far. It improves the 0.5-year exit position error by +93.68% and exit velocity error by +94.01%. Finer stepping actually worsens the isolated exit and 4-year local window. This is the clearest evidence that the trained residual model learned the  $dt=0.08$  encounter correction better than simple finer stepping.

At  $dt=0.10$ , all correction methods help at the 0.5-year exit. Finer-k4 slightly beats EncounterNN on exit-position error, but EncounterNN has the best exit-velocity error. Over the 4-year local window, finer-k2 has the best position time-average, while EncounterNN remains strong and improves position time-average by +51.97%.

## 6. Ensemble Perturbation Results

The ensemble test is important because a single chaotic trajectory can be misleading. Each row below summarizes the final-position gain over ten tiny initial-condition perturbations. Positive values mean improvement over noNN.

**Table 6. Ensemble final-position gain statistics versus noNN.**

| dt   | method      | mean_pos_final_gain_pct | std      | median  | N_positive |
|------|-------------|-------------------------|----------|---------|------------|
| 0.06 | noNN        | +0.00%                  | 0.00%    | +0.00%  | 0/10       |
| 0.06 | EncounterNN | +11.78%                 | 44.35%   | +24.73% | 7/10       |
| 0.06 | finer-k2    | -2.22%                  | 4.97%    | -2.14%  | 4/10       |
| 0.06 | finer-k4    | -2.50%                  | 6.81%    | -2.19%  | 4/10       |
| 0.08 | noNN        | +0.00%                  | 0.00%    | +0.00%  | 0/10       |
| 0.08 | EncounterNN | +3.95%                  | 25.78%   | +2.46%  | 5/10       |
| 0.08 | finer-k2    | -22.60%                 | 108.14%  | +0.75%  | 5/10       |
| 0.08 | finer-k4    | -28.37%                 | 124.36%  | +0.93%  | 5/10       |
| 0.10 | noNN        | +0.00%                  | 0.00%    | +0.00%  | 0/10       |
| 0.10 | EncounterNN | -7.77%                  | 47.40%   | -14.52% | 4/10       |
| 0.10 | finer-k2    | -14.97%                 | 38.63%   | -9.82%  | 5/10       |
| 0.10 | finer-k4    | -1564.88%               | 4646.01% | -34.71% | 4/10       |

At  $dt=0.06$ , EncounterNN has the best ensemble behavior: mean +11.78%, median +24.73%, and improvement in 7 of 10 seeds. Finer stepping is negative on average and positive in only 4 of 10 seeds.

At  $dt=0.08$ , EncounterNN again has the best ensemble mean (+3.95%), but the result is less stable: only 5 of 10 seeds improve and there is a large negative outlier. Finer stepping has positive medians but very negative means because of a severe outlier.

At  $dt=0.10$ , all methods are poor on mean final-position gain. EncounterNN is the least bad by mean (-7.77%), while finer-k2 is worse (-14.97%) and finer-k4 is disqualified due to energy unsafety in one seed. This confirms that  $dt=0.10$  is too aggressive for a clean conclusion.

## 7. How EncounterNN Fared Overall

EncounterNN succeeded at its narrowest intended job: correcting the 0.5-year local encounter exit. It wins the isolated exit-state test at  $dt=0.06$  and  $dt=0.08$ , and is nearly tied with finer-k4 on exit position at  $dt=0.10$  while still winning exit velocity. This means the model is not just adding noise; it learned a real local residual correction.

The strongest EncounterNN operating point is  $dt=0.08$ . It wins the single global 100-year final-position metric, gives a very large local exit-state gain, and remains the ensemble mean winner. This is the best result to highlight if the goal is to show that a learned encounter residual can beat deterministic Zone-3 refinement in the targeted operating regime.

EncounterNN is weaker at  $dt=0.06$  in the global final snapshot. The local exit correction is strong, and the ensemble mean is best, but the single global final position is better under finer-k4. This is not a contradiction: after the encounter, the chaotic system can amplify small differences differently, so the best local correction is not guaranteed to produce the best single final snapshot.

EncounterNN is not reliable at  $dt=0.10$  globally. It helps locally, but the coarse 0.10-year baseline is too aggressive: the global rollout shows EncounterNN worsening all main global metrics. Fine-k2 performs better in the single global run, while fine-k4 becomes energy unsafe. The fair conclusion is that  $dt=0.10$  is a stress test where neither learned correction nor finer stepping is fully dependable.

The practical conclusion is therefore balanced: EncounterNN is a promising learned local correction, not a universal replacement for careful timestep control. It is strongest in the operating region it was designed around, and the results support reporting it as a real but still limited encounter-level surrogate experiment.

## 8. Recommended Interpretation for the Project

Table 7. Recommended conclusions to take forward.

| Point                 | Recommended wording                                                                                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main claim            | EncounterNN can learn a close-encounter residual that simple Zone-3 finer stepping does not always reproduce, especially at $dt=0.08$ .                                            |
| Do not overclaim      | Do not say EncounterNN is globally superior at every timestep. The $dt=0.06$ and $dt=0.10$ global runs are mixed.                                                                  |
| Best evidence         | Use $dt=0.08$ as the clearest success case because it wins global, isolated, and ensemble comparisons.                                                                             |
| Caution evidence      | Use $dt=0.10$ to show why energy safety and timestep limits still matter; k4 fine stepping becomes energy unsafe and EncounterNN is globally weak.                                 |
| Scientific conclusion | Neural residual correction and deterministic fine stepping solve different parts of the problem. Neither alone is a guaranteed replacement for physically constrained integration. |

## Appendix: Source Outputs Used

- comparison\_global\_dt0.06.txt
- comparison\_global\_dt0.08.txt
- comparison\_global\_dt0.1.txt
- comparison\_isolated\_dt0.06.txt
- comparison\_isolated\_dt0.08.txt
- comparison\_isolated\_dt0.1.txt
- comparison\_ensemble\_dt0.06.txt
- comparison\_ensemble\_dt0.08.txt
- comparison\_ensemble\_dt0.1.txt
- summary\_winner\_dt0.06.txt
- summary\_winner\_dt0.08.txt

summary\_winner\_dt0.1.txt

evaluation\_encounter\_fine\_stepping.py